



INDISCRIMINATE UTILIZATION OF SODIUM BENZOATE IN THE PRODUCTION OF COMMERCIALY PROCESSED YOGHURT AND LOCALLY PROCESSED YOGHURT (*KINDIRMO*) IN KANO METROPOLIS, NIGERIA

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ARTICLE INFO.

Key words:

Preservatives, Safety, Additives, Residues. Kano metropolis, Yoghurt

ABSTRACT

The study was conducted to examine indiscriminate use of sodium benzoate (SB) as milk preservative in the commercial and local production of Yoghurt in the study area, and its clinical consequences on public health. A survey was carried out using structured questionnaires among milk wholesalers in Kano metropolis to obtain information regarding the use of SB in yoghurt and *Kindirmo* production, data generated were subjected to descriptive statistics and Analysis of Variance. Also, 15 samples each belonging to both commercially and locally processed Yoghurt were aseptically collected in sample bottles from some selected selling points of commercially processed Yoghurt and *Kindirmo*, and screened for residues of SB using High Performance Liquid Chromatography. The results of the survey revealed that majority of the respondents were young males (52%) and carry out the Yoghurt hawking business outside formal market setting they obtained the raw milk mostly (99%) from Fulani milkmaids and of use SB as preservative without regard to standards set by regulatory agencies (64.50%). The results of the analysis using HPLC indicated that the screened samples contained SB but within the tolerable limit (250 mg/l). However, there are significance difference within the samples. It is concluded that most of the raw milk dealers are using sodium benzoate as preservative, were not aware of the recommended quantity as well as its side effect and residues were detected in products. It is recommended that the use of sodium benzoate as preservative in raw milk should be regulated and monitored by relevant regulatory agencies to safeguard public health.

INTRODUCTION

Food is essential in sustaining life and can be obtained from plants and animals. Milk is one of such foods obtained from mammals' animal (Besthart, 1982) and it has been established as nature's most nutritionally complete food (Everett and McLeod, 2005; Dror and Allen, 2011). The nutritive value of milk depends on its freshness, cleanliness, purity and wholesomeness (Hamid *et al.*, 2003). The domestic supply of milk in Nigeria is about 0.6 million tons whereas demand is 2.0 million tons, leaving a deficit of 1.4 million tons to be filled with imports (FMARD 2016; Price water Coopers, PwC, 2017). The low domestic milk supply is a result of low milk yields due to poor genetic potential of local cattle breeds, poor feeding practices and outdated production practices (FMARD, 2016). Nigeria has an estimated 20 million cattle, of which 2.2 million are used for dairy production and the remainder for meat production (FAOSTAT, 2018). There are three main dairy production systems in Nigeria – the settled Fulani pastoral system, the non-settled Fulani nomadic system, and the large-scale dairy farming system.

Milk is utilized in the production of at least 400 different fermented products all over the world (Okeke, Abdullahi & Makun, 2014). Some of these fermented products indigenous to Nigeria include *Nono*, *Kindirmo*, yoghurt and Cheese. *Kindirmo* is a fermented milk product mostly consumed by Hausa and Fulani tribes in the northern part of the country. *Kindirmo* is full fat or partially-skimmed milk product whereas *Nono* (locally fermented yoghurt) is prepared from skimmed (defatted) milk.

Kindirmo is also used as an important food among the Saharan tribes of West Africa and also popular among the inhabitants of the Mediterranean region and Middle East (Nebedum and Obiakor, 2007). In the Middle East, it is called 'Dahi', 'lassi' or 'Laban' (Nahar, Al-Amin, Alam, Wadud & Islam, 2007). Yoghurt is a semi-solid fermented milk product that originated centuries ago in Bulgaria. It is probably one of the most popular fermented milk products in the world (Niamsiri and Batt, 2009). Its production involves the fermentation of the lactose content in milk giving rise to lactic acid, acetic acid, CO₂, acetaldehyde, diacetyl etc. through the use of starter culture which contains *Streptococcus thermophilus* and *Lactobacillus bulgaricus* (Adolfsson *et al.*, 2004; Tamime and Robinson, 2004, Ifeanyi *et al.*, 2013). Sodium benzoate (E211) is known as the first chemical preservative approved in food products by the US Food and Drug Administration (FDA, 2007). Its solubility is more than other salts of benzoate like potassium and calcium.

There is indiscriminate use of sodium benzoate as preservative in dairy products without regard to regulations set by Standard Organization of Nigeria (SON) and National Agency for Food and Drugs Administration and Control (NAFDAC), and the activities of operators are undertaken under poor hygienic conditions (Ifeanyi *et al.*, 2013). Although sodium benzoate is generally recognized as safe, undesirable effects such as asthma, urticarial, metabolic acidosis and convulsion were observed at low doses in sensitive persons (Hong *et al.*, 2009). Therefore, consumers may be at risk due to indiscriminate and unregulated practices. The risks may include toxicity, damage to body tissues, cancer, food allergy, food intolerance, cancer, multiple sclerosis, attention deficit, hyperactivity disorder, brain damage, nausea, cardiac disease and cardiovascular complications (Ahmed, 2013; Ritu and Rakesh, 2021).

This study was undertaken to examine the presence and concentration levels of sodium benzoate in commercial Yoghurt and *Kindirmo* offered for sale in Kano metropolis.

Materials and Methods

Study Area

The study was carried out in Kano Metropolis which comprises eight local government areas namely Kano Municipal, Fagge, Dala, Gwale, Tarauni, Nasarawa, Ungogo and Kumbotso. Kano State lies between longitude 8° 31' and 0° 19' East and latitude 12° 00' and 0° 43' North in the semi-arid region of Nigeria according to the Global Positioning System (GPS). It is located 481 m above sea level and characterized by tropical wet and dry climate (KNARDA, 2001). Analyses of Yoghurt samples were carried out at the Multi-User Science Research Laboratory of Ahmadu Bello University Zaria.

Survey Study Design

A survey was conducted using a questionnaire. A total of 228 questionnaires were administered for the survey and all the targeted group (milk handlers and yoghurt hawkers)

responded appropriately. A total of 250 yoghurt producers were purposely selected in Kano Municipal, Fagge, Dala, Gwale, Tarauni, Nassarawa, Ungogo and Kumbotso of Kano Metropolis. Information on sodium benzoate usage, status and their experience were documented. All interviews were carried out on the spot at the business premises.

Source of Experimental Materials

Fifteen samples each of commercially prepared Yoghurt and *Kindirmo* study were purchased within Kano Metropolis within Kano Metropolis and used for study.

Preparation of Standards and Samples (Yoghurt and *Kindirmo*)

Individual stock standard solutions of benzoic acid were prepared in deionized water (1000 mg/L) stock standard was in methanol (1000 mg/l). The mixed standard solution was prepared by diluting the stock solution appropriately with mobile phase to give a concentration between 40 mg/L to 120 mg. A mass of 10 g sample was weighted in a beaker and diluted with 35 mL solution of acetic acid 1% and methanol (35: 65 v/v) and then shaken for 15–30 min on stirrer. Then, all content of beaker was transferred to the 50 mL volumetric flask and diluted with extracted solution. Finally, the prepared solution was filtered with 0.45 μ m PVDF syringe filter and was injected to HPLC as described by Fatemeh *et al.*, (2017).

Chromatographic Analysis

Chromatographic analysis was carried out on 1260 Infinity series HPLC from Agilent technology equipped with four solvent lines, high pressure pump, degasser, 20 μ L loop injector, and Ultraviolet detector (UV). The stationary phase is ZORBAX Eclipse-XDB C18 (150 $\times$ 4.6 mm, 3. 5 μ m) and the mobile phase consisted of an aqueous ammonium acetate buffer (pH = 5) and acetonitrile (73: 27 v/v). The separation was performed with a flow rate of 0.7 mL/min and 20 μ L injection volume and by setting the UV detector at 235 nm. The chromatographic system was initially conditioned by the mobile phase for minimum of 1 hour (Fatemeh *et al.*, (2017)

Data Analysis

The data generated from the questionnaires were analyzed using descriptive statistics (percentage and frequency) and Analysis of variance (ANOVA) SPSS statistical version 2.4 at 5% level of probability. Concentrations of SB residues from the Yoghurt samples were generated from the HPLC result.

Results and Discussion

Results of the Survey

Sources of Raw milk and Preservation Methods use by the Respondents

The result in Table 1 shows the sources of the raw milk and preservation methods used by the respondents. It indicated that majority of the respondents obtained the raw milk from Fulani woman while few of them sourced from their personal herds. It was further revealed that most of the respondents were using raw milk for the purpose of making yoghurt and *Kindirmo*. Furthermore, many of them use sodium benzoate to preserve yoghurt and *Kindirmo* products and few rely on refrigeration. It was further revealed that majority of the milk dealers obtained the preservative from local market. It was found out that majority of the milk processors do not measure the quantity of preservatives they use.

Hazard Associated with the use of Chemical Preservatives in Yoghurt and *Kindirmo*

The results in Table 2 shows the level of awareness on set standard and hazard associated with the use of chemical preservatives in the study area. It found out that majority of the respondents were not aware of any standard set by regulatory agencies regarding the use of chemical preservatives as well as negative effect associated with the high used of the

preservatives such as allergy, abdominal upset and vomiting. Majority of the respondents stated that the products could stay up to 8 hours without spoilage when preserved with chemical preservatives. In terms of visitation by regulatory agencies, majority of the respondents were supervised by various agencies such as State Sanitary Officials, local Government Inspectors and National Agency of Food and Drugs Administration and Control (NAFDAC) on quarterly basis.

Means and Standard Errors of Conc. of Sodium Benzoate in Yoghurt and Kindirmo

Concentration of sodium benzoate in the 30 purchased screen samples of yoghurt and *Kindirmo* were presented in Table 3. The result revealed that sodium benzoate was present in all the 30 tested samples of yoghurt and *Kindirmo* respectively. And the samples differ significantly at 5% level. The concentration level recorded in the samples were within the range of 5.45 to 249.25mg/l in yoghurt and 1.01 to 189.32mg/l and 1.01mg/l in *Kindirmo*. However, the ranges recorded in this research did not agreed with work of Mihyar *et al.*, (1999) where the content of benzoic acid were in the range of 10.6 to 1998.8 mg/kg in laban yoghurt in Amman. El-zeiny (2009) also pointed out a high concentration level of benzoic acid of 921 mg/kg in yoghurt in Saudi Arabia. Furthermore, ranges of 9.36 to 26.21mg/l were recorded in 25 yoghurt samples collected from five cities in Turkey (Yildiz *et al.*, 2012). Bahremand and Eskandari (2013) reported that all Doogh samples tested contained sodium benzoate. The work of Akbariadergani *et al.* (2013) were in close agreement with the finding of this research where they reported all of the Doogh samples tested in Iran contained sodium benzoate in the range of 18.3 to 234.1 mg/kg. The reason for the variation in the ranges may be attributed to several factors including the feed of the milk- producing animal, the season of milking, the breed condition, and the content of hippuric acid in the raw milk, type of commercial lactic acid bacteria starter, preservation and processing technique and storage condition (Zahra, Mohammad, Roya & Elham, 2016).

Table 1: Sources of Raw Milk and Preservation Methods Used by the Milk Dealers

Parameters	Frequency	Percentage (%)
Sources of Raw Milk		
Fulani Woman	226	99.12
Others	2	0.88
Total		100
Intended Use of Raw Milk		
Yoghurt	48	21.05
<i>Kindirmo</i>	180	78.95
Total		100
Use of Preservatives in Dairy Products		
Yes	219	96.05
No	9	3.95
Total		100
Means of Preservation		
Refrigeration	66	28.95
Use of Preservatives	162	71.05
Total		100
Preservatives Used for Milk Products		
Sodium Benzoate	210	92.11
Others	17	7.89
Total		100

Use of Sodium Benzoate as Preservative in Yoghurt and Kindirmo	219	96.05
Yes	9	3.95
No		100
Total		
Measurement of Preservative	155	67.98
Yes	73	32.02
No		100
Total		
Instrument Used in Measuring the Quantity of Preservative	177	77.63
Weighing Scale	51	22.37
Assumed Quantity		100
Total	218	95.61
Awareness of Standard Set by Regulatory Agencies	10	4.39
Yes		100
No	54	23.68
Total	30	13.16
	144	63.16
		100
	81	35.53
	147	64.47
		100

Table 2: Hazard Associated With the use of the Chemical Preservatives in Yoghurt and Kindirmo Preparation

Parameters	Frequency	Percentage
i. Awareness of Health Hazard Associated with the use of Chemical Preservatives		
Yes	15	6.58
No	213	93.42
ii. Effect of Using Chemical Preservatives		
8 Hours		83.33
12 Hours	190	15.35
16 Hours	35	1.32
iii. Shelf life of Chemically Preserved Yoghurt		
8 Hours		47.37
12 Hours	108	39.91
16 Hours	91	14.91
iv. Shelf life of Chemically Preserved <i>Kindirmo</i>		
8 Hours	93	40.79
12 Hours	51	22.37
20 Hours	34	14.91
48 Hours	50	21.93
v. Monitoring Visit Regulatory Agencies		
Yes		91.23
No	208	8.77
vi. Regulatory Agency		
vii. National Agency for Food, Drugs Administration and Control	6	2.63
viii. Standard Organization of Nigeria	1	0.44
ix. State Sanitary	166	72.81
x. Local Government Health Inspector	35	15.35
Total		100
Frequency of Visit		
Monthly	6	2.63
Quarterly	188	51.75
Annually	94	41.23
Total		100

Table 3: Means and Standard Errors of Conc. of Sodium Benzoate in Yoghurt and *Kindirmo*

Sample Number	Conc. of SB in Yoghurt	p-Value	Conc. of SB in <i>Kindirmo</i>	p-Value
1.	159.79±133.80 ^a	0.031054*	3.06±15.23 ^c	0.02341*
2.	23.39±15.23 ^f		1.02±15.23 ^c	
3.	154.16±15.23 ^a		1.94±15.23 ^c	
4.	34.96±15.23 ^e		3.3±15.23 ^c	
5.	69.28±15.23 ^d		1.22±15.23 ^c	

6.	8.69±15.23 ^f	4.12±15.23 ^c
7.	47.51±15.23 ^e	1.74±15.23 ^c
8.	127.69±15.23 ^a	188.64±15.23 ^a
9.	249.25±15.23 ^a	3.4±15.23 ^c
10.	16.56±15.23 ^f	22.24±15.23 ^b
11.	5.45±15.23 ^{fg}	3.09±15.23 ^c
12.	104.68±15.23 ^b	27.5±15.23 ^b
13.	82.65±15.23 ^c	1.66±15.23 ^c
14.	100.93±15.23 ^b	2.14±15.23 ^c
15.	18.83±15.23 ^f	3.44±15.23 ^c

SB = Sodium benzoate, * = p<0.05, p-Value = Probability value s

Conclusion

The experiment revealed that sodium benzoate was present in all the samples of commercially processed yoghurt and *kindirmo* screened samples sold in Kano metropolis. The value of and SB detected in the commercial yoghurt (249 mg/L) and *Kindirmo* (189 mg/L) offered for sale in the study area in the study were within the recommended levels of SB preservative approved by both Nigerian industrial standard (1997) and Codex Alimentarius (2003). Finally agency of government in charge of food and drug regulation and control should continue to monitor the use of sodium Benzoate at recommended levels (250mg/l) in processing and preserving dairy products in the study area.

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